

EduGenie – Development Plan

Project: EduGenie – Google Gemini Powered Learning Assistant

1. Introduction

The development plan describes how the EduGenie application will be developed and organized from the initial design stage to the final working system.

2. Development Activities

2.1 Backend Development

Develop the FastAPI backend to receive user requests and manage communication between the frontend and learning modules.

2.2 Learning Module Development

Develop separate modules for:

- Question Answering
- Simplified Explanation
- Quiz Generation
- Text Summarization
- Learning Path Recommendation

2.3 AI Integration

Integrate the Google Gemini API to provide AI-powered responses for the required learning functions.

2.4 Frontend Development

Develop a simple HTML and CSS interface where students can select a task, enter their requirements, submit requests, and view results.

2.5 API Integration

Connect the frontend with the FastAPI backend through appropriate endpoints.

2.6 Error Handling

Implement suitable error handling for invalid inputs, API failures, and configuration problems.

2.7 Testing

Test each major feature individually and then test the complete application to ensure that the different components work together correctly.

2.8 Deployment

Prepare the application for deployment so that the EduGenie system can be accessed through a web browser.

3. Development Approach

The project follows a structured development approach:

**Planning → Design → Development → Integration → Testing → Deployment
→ Demonstration**

4. Expected Outcome

The final system should provide students with a web-based AI learning assistant that can answer questions, explain concepts, generate quizzes, summarize text, and provide learning recommendations.

5. Conclusion

This development plan provides a clear sequence of activities for building and completing the EduGenie project while keeping the different components organized.